

DINESH KUMAR M

Software Engineer | Backend Engineer

Java · Spring Boot · Rust · Microservices · Distributed Systems

Hyderabad, Telangana, India (Open to Relocation) | +91-97879-43467 | mdkdinesh2503@gmail.com

LinkedIn: [linkedin.com/in/mdkdinesh2503](https://www.linkedin.com/in/mdkdinesh2503) | GitHub: github.com/mdkdinesh2503 | Portfolio: mdkdinesh2503.netlify.app

PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 2 years of experience designing, developing, deploying, and supporting production backend systems for a gaming platform serving 20K+ registered users. Experienced in developing Java monolithic applications and Rust microservices, delivering scalable backend solutions through REST APIs, GraphQL, gRPC, database design, distributed service communication, caching, and production support. Hands-on experience across the SDLC, including requirement analysis, solution design, implementation, deployment validation, production support, incident resolution, and Agile collaboration.

TECHNICAL SKILLS

Languages: Java · Rust · TypeScript · JavaScript · SQL

Backend: Spring Framework · Spring Boot · Spring MVC · Spring Data JPA · NestJS · Node.js

Frontend: Next.js · React · Angular

API Technologies: REST APIs · GraphQL · Apollo GraphQL · gRPC · Protocol Buffers

Databases: PostgreSQL · Redis · DynamoDB · MySQL · MongoDB · NeonDB

Backend Engineering: API Design · Database Design · Data Modeling · Distributed Systems · Microservices · Monolithic Architecture · Event-Driven Architecture · SQL Query Optimization · Redis Caching · Authentication · Authorization · RBAC · Performance Optimization · Production Support · Root Cause Analysis · System Design

DevOps & Developer Tools: Git · Docker · Jenkins · ArgoCD · Grafana · Maven · Postman · Altair GraphQL · RedisInsight · pgAdmin · NoSQL Workbench · Termius · Warp

PROFESSIONAL EXPERIENCE

Aretedge Innovations Pvt. Ltd.

Associate Software Engineer

Hyderabad, Telangana, India (On-site)

Aug 2024 – May 2026

Domain: Gaming Platform | **Architecture:** Java Monolith → Rust Microservices

Tech Stack: Java · Spring Boot · Rust · NestJS · PostgreSQL · Redis · DynamoDB · GraphQL · gRPC · Docker · Jenkins · ArgoCD · Next.js · Grafana

Feature Ownership

- Owned the end-to-end backend implementation of the Challenge Module, translating business requirements into production-ready backend workflows through API design, database modeling, business logic implementation, deployment validation, and production support.
- Designed and implemented backend services for the Casual Games Module, integrating external gaming provider APIs and implementing gameplay, participation, rewards, and leaderboard workflows aligned with business requirements.
- Designed and implemented Double Elimination tournament workflows, translating complex tournament rules into maintainable backend logic that became the foundation for tournament functionality.

Backend Engineering

- Designed and delivered 20+ production backend APIs across Java monolithic applications and Rust microservices using REST APIs, GraphQL, and gRPC while implementing JWT-based authentication, RBAC, and distributed service communication.
- Developed GraphQL schemas and resolvers, gRPC services, and Protocol Buffer definitions to enable efficient communication between distributed backend services.

Database Engineering

- Designed PostgreSQL schemas, indexes, JSONB structures, materialized views, and database functions while optimizing SQL queries for scalable transactional gaming workflows.
- Designed DynamoDB data models by defining partition keys, sort keys, and Global Secondary Indexes (GSIs) based on application access patterns while following established architectural conventions.
- Developed Node.js migration scripts to migrate application data from DynamoDB to PostgreSQL, supporting schema redesign and data transformation during backend modernization.

Performance & Scalability

- Implemented Redis caching strategies using TTL and database fallback mechanisms to reduce repeated database access for leaderboard and other high-read workflows while maintaining data consistency and improving backend performance.

Production Engineering

- Supported 25+ production releases, resolved 10+ production incidents, and delivered 8+ production hotfixes by analyzing ArgoCD service logs, performing root cause analysis, validating fixes through API testing, and coordinating deployments using Jenkins with QA and DevOps teams.

Engineering Collaboration

- Participated in architecture discussions, API and database design reviews, sprint planning, code reviews, technical documentation, and the SDLC while collaborating with architects, QA, DevOps, and product stakeholders.

INTERNSHIP & TRAINING

Aspire Systems

Graduate Engineer Trainee

Graduate Engineering Intern

Graduate Engineer Trainee

Chennai, Tamil Nadu, India (Remote)

Apr 2023 – Jul 2024

Jan 2023 – Mar 2023

Sep 2022 – Dec 2022

Training Focus: Enterprise Software Engineering

Tech Stack: Java · Spring Boot · Angular · Node.js · MySQL · REST APIs · Git

- Completed enterprise software engineering training in Java, Spring Boot, Angular, Node.js, MySQL, object-oriented programming, software architecture, and SDLC through project-based development.
- Built enterprise-style full-stack applications by implementing frontend features, backend business logic, relational database operations, debugging, API testing, and issue resolution following enterprise development practices.
- Applied object-oriented programming, layered architecture, modular application design, relational database design, version control, Agile development practices, and collaborative software engineering principles through project-based assignments.

PROJECTS

Personal Finance Tracker | *Next.js · TypeScript · PostgreSQL · NeonDB · Zustand*

Links: [Live Demo](#) | [GitHub](#)

- Designed, developed, and deployed a personal finance management application implementing secure authentication, Row-Level Security (RLS), responsive dashboards, and API-driven workflows for expense tracking and financial analytics.
- Improved application responsiveness by replacing a Supabase + Drizzle ORM implementation with a simplified Neon PostgreSQL architecture after evaluating authentication latency and architectural trade-offs.

Student Management System | *Java · Angular · MySQL*

Links: [Live Demo](#) | [GitHub \(Angular\)](#) | [GitHub \(Java\)](#)

- Designed and developed a role-based academic management platform implementing REST APIs, relational database design, and RBAC to support attendance, examinations, fee management, and administrative workflows.

E-Commerce Platform | *Spring Boot · Spring Data JPA · MySQL*

Links: [Live Demo](#) | [GitHub \(Frontend\)](#) | [GitHub \(Spring Boot\)](#)

- Designed and developed a layered Spring Boot backend using REST APIs, Spring Data JPA, and relational database modeling to support modular product, cart, and order management workflows.

Travel Agency Platform | *Angular · Node.js · MongoDB*

Links: [Live Demo](#) | [GitHub](#)

- Designed the architectural foundation for a modular travel management platform by implementing authentication, backend services, and MongoDB-backed data models.

EDUCATION

K.S.R College of Engineering

Bachelor of Engineering (Computer Science and Engineering)

Namakkal, Tamil Nadu, India

Aug 2019 – May 2023

CERTIFICATIONS

AWS DynamoDB Fundamentals – Meta | **Rust Programming Essentials – Edureka** |

Advanced React – Whizlabs | **Google Prompting Essentials – Google**